

MLE5217 – Take Home Assignments

Important!

There is no one correct answer.



You will be evaluated based on **how you critically analyse the materials science problem, make decisions on how to proceed with the task and explain your decisions and thinking!**

Important! : *#Comments*

- Write clear comments in the code so that another person can follow the logic.
- In instances where you have made decisions - justify them.
- Whenever, you have attempted more than one possibility, leave the original attempts (if possible) - don't delete or overwrite (you may comment them if needed).
- Acknowledge (if any) references or AI tools used at the bottom of the notebook.

Evaluation (Note that the allocated marks are tentative)

The primary emphasis will be on the depth and thoroughness of your approach to the problem. Key areas of focus will include:

- * Data Exploration: Demonstrating a thorough investigation of the data, exploring different analytical possibilities, and thoughtfully selecting the best course of action. (~7 marks)
- * Implementation: Translating your chosen approach into clean and efficient code. (~5 marks)
- * Machine Learning Process: Executing the machine learning process correctly and methodically, ensuring proper data handling, model selection, and evaluation. (~5 marks)
- * Clarity of Explanation: Providing clear explanations of each step, with logical reasoning for the decisions made. (~10 marks)
- * Critical Analysis: Identifying any limitations of the approach, suggesting potential improvements, and making relevant statistical inferences based on the results. (~8 marks)

Submission

Ensure that

- each of the cells of code in the final Jupyter notebooks have been Run for output (Except for the hyper-parameter optimization if any).
- Name the file by your name as "YourName.ipynb".
- It is your responsibility to Ensure that the correct files are being submitted, and the file extensions are in the correct format (.ipynb).
- Submission will be via Canvas, and late submissions will be penalized.

Assignment 2 (35 marks)

Building a Classification Model to distinguish between metallic & non-metallic compounds based on bandgap data.

Dataset: Assign_classification_data.csv

The data contains the chemical formula and energy band gaps (in eV) of experimentally measured compounds. These measurements have been obtained using a number of techniques such as diffuse reflectance, resistivity measurements, surface photovoltage, photoconduction, and UV-vis measurements. Therefore, it is acceptable for a given compound to have multiple values in the dataset.

Objective

Fit a suitable Classification model to separate metals from non-metals in the data.

Points to note:

- Follow the general machine learning process
- Try different composition-based feature vectors to extract information from the given data composition before deciding on the best featurization scheme.
- Additional step (not mandatory): If you think is necessary, you may choose to supplement the data by adding in any other materials data as descriptors (features). This can be done via either an online database such as Materials Project or using commercial AI assistants.
- You may use your judgement on how to differentiate (bandgap threshold) between metals & non-metals.
- Use suitable metrics to quantify the performance of the classifier.
- For added advantage you may optimize the hyper-parameters of the model. Note: Optimization algorithms can require high processing power, therefore may cause your computer to freeze (Ensure you have saved all your work before you run such codes). In such a case you may either do a manual optimization or leave the code without execution.
- Comment on the overall performance of the model.

